

Campaign ROI & Media Spend Simulator - Recommendation Memo

Why this project

Built for the Tata CLiQ Monetization Intern application, whose two stated goals are optimizing revenue streams and improving advertiser ROI - measured on pitch conversion, new revenue, and campaign renewals. This is, literally, the tool a retail-media team runs before greenlighting spend: at what point does a campaign clear its own cost.

Methodology

1. **Benchmark sourcing** (data/raw/ad_benchmarks.csv) - CTR, CVR, CPC from WebFX's 2026 Google Ads Benchmarks (global) and OwlClaw's PPC Benchmarks India 2026 (India-specific search/shopping figures); India blended AOV from a 2026 order-level analysis; USD:INR from the Fed's H.10 release.
2. **CPM derivation** - not directly disclosed by any source, so derived as $\text{CPC} \times \text{CTR\%} \times 1000$ and shown as a formula in both the Python model and the Excel workbook, never asserted as a benchmark.
3. **Funnel** - Spend -> Impressions -> Clicks (CTR) -> Orders (CVR) -> Gross Revenue (AOV) -> Incremental Revenue (haircut) -> Incremental ROAS.
4. **Break-even solver** - $\text{fixed_cost} / (\text{incremental_ROAS} - 1)$ when $\text{ROAS} > 1$; explicitly returns "NEVER" when $\text{ROAS} \leq 1$, rather than a misleading negative or infinite number.
5. **Sensitivity tornado** - CTR, CVR, and incrementality haircut each moved $\pm 20\%$, the three assumptions most likely to move a real decision.

Results

At INR 500,000 test spend: Display returns 0.9430x incremental ROAS (never breaks even at any spend level under these assumptions); Shopping/Sponsored returns 4.2908x (breaks even at just INR 22,791 of spend, derived from full-precision ROAS - matches the Excel workbook's live-formula result exactly). Full sensitivity table: [outputs/tables/funnel_and_sensitivity.csv](#).

What I learned

CTR and CVR sensitivity produce *identical* ROAS deltas (± 0.1886 for a $\pm 20\%$ move in either lever) - not a bug, but a direct consequence of the funnel being multiplicatively linear in both. Worth stating plainly: in this model shape, which lever "matters more" isn't a modelling question - it's whichever lever is operationally cheaper to move in a real campaign (creative/targeting for CTR vs. landing-page/offer for CVR).

Cross-validation

The Excel workbook (excel_model/Campaign_ROI_Simulator.xlsx) mirrors the Python model's logic as live formulas referencing an Assumptions sheet - no pasted values anywhere in Funnel_Model or Sensitivity. Recalculated headlessly via LibreOffice and compared line-for-line against the Python output: both agree to the cent (Display 0.9430x, Shopping 4.2908x). This is the same build-two-independent-paths-and-reconcile discipline used elsewhere in the portfolio.

Limitations

See LIMITATIONS.md. In short: CPM is derived, not disclosed; fixed cost and incrementality haircut are stated illustrative assumptions; benchmarks are national/global averages, not category-specific or Tata-CLiQ-specific.

What this demonstrates for the role

Direct evidence of the exact analytical tool the JD describes - a break-even/ROAS model built to Excel-fluency standard (live formulas, not pasted values) - plus the honesty to report a channel that doesn't work under real assumptions rather than only showing the flattering number.